

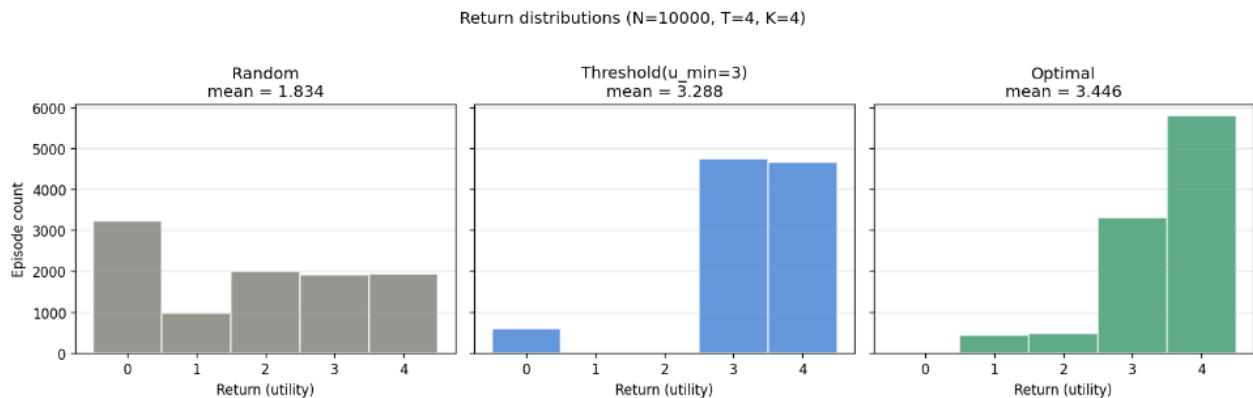
## Problem 3 -- Gymnasium implementation and benchmarks

ApartmentEnv,  $T = K = 4$ ,  $N = 10,000$  episodes per condition.

### (c) Empirical comparison [12 pts]

Results at  $\sigma = 0$ :

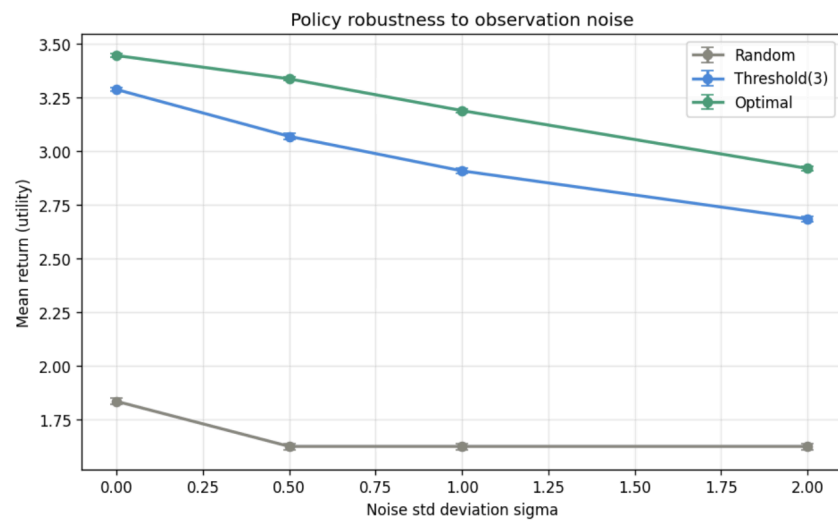
| Policy                            | Mean utility | SE    | Frac. reject-all |
|-----------------------------------|--------------|-------|------------------|
| RandomPolicy                      | 1.834        | 0.015 | 0.322            |
| ThresholdPolicy( $u_{\min} = 1$ ) | 2.514        | 0.011 | 0.000            |
| ThresholdPolicy( $u_{\min} = 2$ ) | 2.981        | 0.008 | 0.003            |
| ThresholdPolicy( $u_{\min} = 3$ ) | 3.288        | 0.010 | 0.060            |
| ThresholdPolicy( $u_{\min} = 4$ ) | 2.743        | 0.019 | 0.314            |
| OptimalPolicy                     | 3.446        | 0.008 | 0.000            |



Return distributions at  $\sigma = 0$ .

**Discussion.** Best fixed threshold is  $u_{\min} = 3$  (mean 3.29). Optimal beats it at 3.45, matching theoretical  $W_1 = 3.44$  from Problem 1(c) -- confirms correct implementation. No fixed threshold matches optimal because it cannot adapt to the remaining horizon.

### (d) Robustness to noise [8 pts]



Mean utility vs. observation noise sigma.

**Discussion.** Random is flat (ignores observations). Threshold(3) degrades fastest (-0.61 at sigma = 2). Optimal is most robust (-0.53), saved by its week-4 always-accept rule. As sigma grows, the signal becomes useless and policies converge toward week-only rules, consistent with Problem 1(d)(iii).